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Next item

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1. Which of the following statements are true about K-fold Cross-Validation? 0.5 / 1 point

- [x] It eliminates the need for a separate validation set.

 This should not be selected
 Incorrect. K-fold Cross-Validation still requires splitting the data into training and validation sets, but the process is repeated multiple times.

- [x] It minimizes the bias in the performance estimate compared to a single train-test split.

 Correct
 Right! K-fold Cross-Validation reduces bias as every observation is used for both training and validation.

- [x] It helps in assessing the model's ability to generalize.

 Correct
 Correct! K-fold Cross-Validation provides a good estimate of model performance on unseen data.

- [x] It ensures that the training and validation sets are completely independent.

 This should not be selected
 Not quite. K-fold Cross-Validation involves repeated sampling, so the same data points can be in both training and validation sets across different folds.

2. Which of the following statements about underfitting in machine learning models are correct? 1 / 1 point

- [] Data augmentation is a common technique used to address underfitting.
- [] Underfitting typically results in high training error but low validation error.
- [x] A model with high bias is likely to underfit the data.

 Correct
 Correct! High bias in a model often leads to underfitting as the model is too simplistic to capture the data's complexity.

- [x] Underfitting occurs when a model is too simple to capture the underlying patterns in the data.

 Correct
 Correct! Underfitting is usually the result of a model that is too simple or lacks sufficient complexity to capture the data's patterns.

- [x] Underfitting can often be addressed by adding more features to the model or increasing its complexity.

 Correct
 Correct! Increasing the model's complexity or adding more features can help address underfitting by allowing the model to capture more of the underlying data patterns.

- [] Underfitting can be diagnosed by evaluating the model's performance on the test set alone.

3. Which of the following steps are part of performing a complete backward pass in a neural network? 0.8333333333333334 / 1 point

- [x] Calculate the dot product of the input and weights

 This should not be selected
 Incorrect. Calculating the dot product of the input and weights is part of the forward pass, not the backward pass.

- [x] Compute the derivative of the activation function

 Correct
 Correct! Computing the derivative of the activation function is an important part of the backward pass.

- [] Preprocess and scale training data
- [x] Calculate the gradient of the loss with respect to each weight

 Correct
 Correct! Calculating the gradient of the loss with respect to each weight is a crucial step in the backward pass.

- [] Initialize weights and biases
- [x] Update the weights using the calculated gradients and the learning rate

 Correct
 Correct! Updating the weights using the gradients and learning rate is essential during the backward pass.

4. Which of the following tasks are essential steps in implementing a neural network from scratch using NumPy? 0.8 / 1 point

- [x] Initializing weights and biases

 Correct
 Correct! Initializing weights and biases is a critical step in building a neural network.

- [x] Implementing the forward and backward passes

 Correct
 Correct! Implementing both forward and backward passes is essential for training the neural network.

- [x] Preprocessing and scaling training data

 This should not be selected
 Try again. While preprocessing is important, it is not part of the core steps when implementing a network from scratch using NumPy.

- [] Using pre-built functions from deep learning libraries like TensorFlow
- [x] Calculating the dot product between input and weight vectors

 Correct
 Great job! Calculating the dot product is crucial in the forward pass of the neural network.

5. Which of the following best describes the role of the optimizer in neural network training? 1 / 1 point

- () It computes the loss function
- (•) It updates the weights and biases using the calculated gradients
- () It adjusts the learning rate during training
- () It normalizes the input data

 Correct
 Well done! The optimizer uses the gradients calculated during backpropagation to update the weights and biases, thereby minimizing the loss.